Lab 01: Linear Regression Project

1 General Regulations

- The assignment must be completed in registered groups.
- Any member who does not participate will receive no points for this assignment.
- The work must comply with the project requirements.
- All reference materials (if any) must be fully cited in the "References" section of the report.
- Support tools such as ChatGPT, GitHub Copilot, etc., may only be used as references; the content must be reviewed and adjusted to fit the given problem. If it is found that AI tools were used excessively or that the content generated is inappropriate/incorrect, up to 50% of the score may be deducted (depending on the severity).
- Submissions that are identical, plagiarized, or copied from existing works will result in **a zero grade for the entire course**.

2 Project Description

In this project, you will work on a Regression problem by applying Linear Regression models to analyze and predict a continuous target variable. You are encouraged to choose your own problem based on a real-world dataset or a dataset of your interest. The goal is to apply the fundamental concepts of regression modeling and analytical reasoning to understand and explain the relationship between variables.

3 Project Requirements

3.1 Problem Selection and Definition

- You must select a regression problem of your own choice.
- Clearly explain the motivation behind choosing this problem — why it is interesting, relevant, or useful.

- Define your problem objective precisely (e.g., predicting housing prices, estimating car fuel efficiency, forecasting student grades, etc.).
- Clearly state what you want your model to predict and what variables (features) are used as inputs.

3.2 Model Construction

- You are required to apply Linear Regression as your main modeling approach.
- You may design and compare multiple Linear Regression models.
- You must justify your design decisions: explain why you selected certain variables, transformations, or model variants, and how these choices align with your objectives.
- Each group must develop at least as many Linear Regression models as the number of students in the group

3.3 Model Evaluation and Interpretation

- Evaluate your models using appropriate regression metrics (e.g., MAE, MSE, R^2, ...).
- Compare the performance of the different models you built.
- Interpret your results strictly based on the model outcomes. You should not assume external domain knowledge or unrelated information outside what is reflected by your model results.
- Discuss what the coefficients and performance metrics reveal about your data and model.

3.4 Application Development

- Develop a simple application or interface (e.g., a web app, desktop app, or notebook-based interface) that allows users to:
  - Input new data or adjust feature values to generate predictions from your trained regression model.
  - Display prediction results in a clear and intuitive format.
  - Visualize key trends and relationships in the data (e.g., through charts, plots, or dashboards) to help users understand how the input variables influence the predicted output.
  - Optionally, provide summary statistics or performance insights (e.g., R^2, RMSE) to communicate the model's effectiveness.

4 Submission Format

- Each group must designate one representative to submit the assignment.
- The submission must be a compressed file named **Group [Group-ID].zip**, which includes:
  - A folder containing the dataset or a link to the dataset.
  - A folder containing the source code or Github link.
  - An environment file (e.g., **requirements.txt** or **environment.yml**) listing all the libraries and dependencies required to run the project.
  - A video that demo our application.
  - Your final report (PDF format, 10-20 pages recommended) should include:
    * Problem description and motivation
    * Data description (source, preprocessing, visualization)
    * Model design and explanation
    * Evaluation and comparison of results
    * Conclusion and insights drawn from the model outcomes
    * Brief description of your application, including its main features, user interface, input/output functionality, and how it integrates the trained model.
    * References
- If the dataset is too large, you may upload it to an external server such as Google Drive and submit the public link (which must remain accessible for at least two years). The code and report files must still be submitted on Moodle.
- If the dataset or source code link is inaccessible, 50% of the total score will be deducted.

5 Contact Information

For any questions or issues during the project, please send an email to vntan.work@gmail.com